Kabeen Kim

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Research Interests: *Faithful RAG · Multi-hop GraphRAG · Knowledge Graphs · Evaluation & Error Analysis of LLM Systems*

Education

Duksung Women's University — *Seoul, South Korea*

B.E. in Computer Engineering & B.S. in Statistics | GPA: 4.15 / 4.5 (3.85 / 4.0 U.S. equivalent) — *Mar 2021 – Feb 2027 (Expected)*

University of Mississippi — *Mississippi, USA*

Exchange Student, Computer Science — *Aug 2024 – Dec 2024*

Publications

[1] Haopeng Chen, Yihao Ai, **Kabeen Kim**, et al. "UDAPose: Unsupervised Domain Adaptation for Low-Light Human Pose Estimation." *Conference on Computer Vision and Pattern Recognition (CVPR), 2026.* (Poster) [Paper] [Code]

[2] **Kabeen Kim***†, Yena Kim*, Minjeong Seo*. "ConFuse: Context-Aware Fusion of LLMs and GBDTs for Sleep Prediction from Lifelog Data." *International Conference on ICT Convergence (ICTC), 2026.* (Poster) [Paper] [Code]

[3] **Kabeen Kim**†, et al. "CoDU: Multi-Stage Approach for Complex Document Structure Understanding." (in Korean) *Korean DataBase Conference (KDBC), 2025.* (Oral) [Paper] [Code]

[4] **Kabeen Kim**†, et al. "SPTC (Single Pass Tree Chain-of-Thought): An Efficient Prompting Scheme for Small Language Models." *Korea Computer Congress (KCC), 2025.* (Poster) [Paper] **Equal contribution. †Presenter.*

Research Experience

Electronics and Telecommunications Research Institute (ETRI) — *Daejeon, South Korea*

Trainee of Researcher, Government-Funded Research Institute — *Jul 2026 – Aug 2026*

Multi-hop GraphRAG – *Tracing Evidence Loss over a Knowledge Graph*

- Built a ∼**38K-relation** knowledge graph from a technical manual and **250 multi-hop questions** to trace evidence loss.
- Rebuilt traversal and reranking to score whole paths, retrieving all required evidence for **202 of 250 questions** (from 62), over twice the strongest baseline.
- Reached **Recall@5 of 0.873** on 300 HotpotQA questions, 2nd among compared methods behind HippoRAG (0.898).
- Built an LLM judge scoring correctness, faithfulness, completeness, and provenance. Evidence grounding raised **correctness 2.84 → 3.86/5**, adopted in a lab project.

Data Mining & Knowledge Discovery Lab, Duksung Women's University — *Seoul, South Korea*

Undergraduate Researcher — *Mar 2025 – Feb 2026*

SPTC [4] – *Efficient Prompting for Small Language Models*

- Proposed and led a first-author study on whether one prompt can merge generation, self-evaluation, and selection.
- Ran **1.5× faster than ToT** at 1.4 pp lower accuracy and **21.3 pp above CoT**, with errors traced to self-evaluation.

CoDU [3] – *2025 Samsung AI Challenge, 3rd of 210 Teams*

- Recruited and led a 3-person team, designing the layout detection, OCR, and reading-order pipeline.
- Labeled 322 documents to fine-tune DocLayout-YOLO (**F1 0.463 → 0.580**), which also raised downstream OCR and reading-order scores without modifying those modules.
- Parallelized detection and OCR for the **fastest inference** in the competition.

Product Search RAG – *Lab Collaboration*

- Defined the evaluation, then tuned retrieval and reranking against it: **F1 0.3 → 0.6**, leading to a follow-up contract.

Vision and Multimodal Intelligence Lab, University of Mississippi — *Mississippi, USA*

Undergraduate Researcher — *Sep 2024 – Aug 2025*

UDAPose [1] – *Unsupervised Domain Adaptation for Low-Light Pose Estimation*

- Joined via faculty outreach and sorted baselines' low-light failures into **15 categories**, many in dark or cluttered scenes.
- Ran domain adaptation experiments on these failures (**+10.1 AP**), continuing remotely in biweekly meetings from Korea.

Work Experience

Boston Consulting Group (BCG) *Seoul, South Korea*

Research Analyst *Mar 2026 – Jun 2026*

Domain LLM & RAG QA – *Fine-Tuning, Evaluation, and Slide Parsing*

- LoRA-tuned Qwen models for domain QA, raising **LLM judge scores 2.3 → 3.3/5**, with gains only in the largest model.
- Automated QA-pair checks (deduplication, grounding) and built an evaluation framework (EM, token F1, LLM judge).
- Built a slide parser and value-extraction agent, **ranking 1st of tested parser–LLM pairs** with Claude Sonnet 4.6.

Selected Projects

ConFuse [2] – Context-Aware Fusion of LLMs and GBDTs *github.com/kbannie/ConFuse*

Research Team Lead, ToBig's Big Data & AI Society *May 2026 – Sep 2026*

- Led a team fusing GBDTs and LLMs for sleep prediction from small lifelog data, fine-tuning **Qwen2.5-7B with QLoRA**.
- Weighted the LLM per sample by input quality, reaching **0.6415 log loss**, below ML-only and 50:50 fusion.

Dimension of Memory – Emotion-to-Music Immersive Exhibition *github.com/4CO4CO/4co4co-ai*

AI Lead, 4-Person Capstone Team – Hanium Dream-Up (Government Mentoring Program) *Feb 2025 – Nov 2025*

- Led single-GPU serving within budget, choosing models by measured performance.
- Built a CLIP-based **17-emotion classifier** for photos without people using ∼6K new scenes and evaluated it per emotion.
- Exhibited the system for 3 days, drawing **95% visitor satisfaction**.

Honors & Awards

2026 **VOOY Track Award**, OBA RUN 2nd Hackathon
2025 **3rd Place (out of 210 teams)**, Samsung AI Challenge – Visually-Rich Document Understanding
2024 **31st Mirae Asset Overseas Exchange Scholarship (∼USD 5K)**, Mirae Asset Park Hyeon Joo Foundation
2024 **Excellence Award**, TAVE 13th Data Analysis Project
2024 **2nd Prize**, DS Global Vision Makers Innovation Idea Competition
2021–22 **Academic Excellence Scholarship (3 awards)**, Duksung Women's University

Leadership

Duksung Global Challenger

Team Lead & Project Proposer, "Be a Connector" (WeCon) *Mar 2025 – Sep 2025*

- Secured ∼**USD 7K in funding** to build a BLE beacon app linking the campus with local businesses.
- Interviewed experts in 4 European countries in English on BLE/UWB tradeoffs, shaping app features and budget.

Teaching & Volunteering

Chungnam National University & NAQS (Korean Government Agency)

Teaching Assistant, n8n & RAG Workshop *Sep 2025 – Oct 2025*

- Authored **170 slides** and live demos on API authentication, Docker, and n8n for 20-hour lectures (50 participants).
- Prototyped a RAG QA lab and **3 workflow automation labs** with n8n for public-sector staff.

CJ OliveNetworks SW Creative Camp

University Volunteer Instructor, CJ UNIT (8th Cohort) *Mar 2023 – Jun 2023*

- Taught **40 hours of LEGO robotics and Python coding** to middle school students in Jeju via remote classes.
- Co-designed a 3-session ESG curriculum on wave, solar, and wind energy with LEGO guides and Python examples.

Skills

- **Programming & Statistics**: Python, R, C++, SQL (SQLD certified), regression, time series, multivariate analysis
- **ML & Deep Learning**: PyTorch, Unsloth (LoRA/QLoRA), OpenCV, YOLO, XGBoost, LightGBM
- **LLM & Retrieval**: LangChain, Qdrant, Pinecone, NetworkX
- **Infrastructure & Tools**: PostgreSQL, AWS (EC2/S3), Docker, Git, Weights & Biases, n8n